

2023



Securely Migrating Gas Network Infrastructure Using AWS with SGN

Learn how gas distribution company SGN securely migrated critical infrastructure to the cloud using AWS Professional Services.

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Increased flexibility

Created

Enhanced





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to create new testing environments

opportunities to innovate

security and compliance

Delivered

strong network reliability

Overview

Gas distribution company [SGN Ltd.](#) (SGN) was looking to migrate its gas network infrastructure to the cloud and needed a provider that it could work closely with due to the highly regulated nature of the gas industry. SGN turned to Amazon Web Services (AWS), having previously migrated its corporate infrastructure to AWS. By migrating its critical gas network infrastructure to AWS, SGN achieved increased flexibility to create new testing environments, created opportunities to innovate, enhanced security and compliance, and delivered strong network reliability.

Opportunity | Using AWS to Modernize Gas Network Infrastructure for SGN

SGN is a UK gas distribution company that manages the natural and green gas networks of nearly six million homes and businesses across Northern Ireland, Scotland, and the south of England. SGN manages 74,000 km of gas mains along the network and delivers emergency services for gas safety and leakage. SGN strives to maintain high standards of customer service and reliable reporting to stakeholders and regulators while maintaining a cost-effective, data-driven operation.

In 2015, SGN was looking for ways to adapt for the future. The company wanted to explore opportunities to migrate from legacy on-premises data centers to the cloud. Over the next 5 years, SGN underwent a transformation program to migrate its corporate infrastructure to the cloud, successfully migrating 99 applications securely to AWS. In conjunction with the corporate migration, the company also built data lakes and hosted analytics on AWS. In 2019, SGN decided to tackle the even more challenging task of migrating its gas control operations to the cloud. The company's gas network infrastructure was nearing the end of its life cycle and needed to be refreshed to comply with gas control regulations. This project would be a virtually unprecedented all-in-cloud migration due to the scale and complexity of the infrastructure.

"We had to deal with the security threats that come with aging infrastructure, but replacing the physical infrastructure meant a physical move as well," says Arun Rajaram, program management consultant at SGN. "This was a key driver for us to choose AWS and continue our journey to the cloud."



Solution | Meeting High Security and Compliance Standards Using AWS Professional Services

There were many security and compliance considerations for a cloud migration of this scale. SGN's infrastructure is classed as Critical National Infrastructure, which is regulated by guidelines set out by the National Cyber Security Centre. The company would need to deliver several presentations to the National Cyber Security Centre to show proof of concept and demonstrate that the applications to support gas control were sufficiently reliable on the cloud. For its successful corporate migration, SGN had already been working closely with the AWS team and felt confident consulting with them on how best to navigate the gas control phase. "We carried a challenging project over the line by bringing in the best people for the job," says Charlie Smith, head of IT Project and Programme Delivery at SGN. "They had the skills and knowledge of the tooling to get it done right."

For its gas control migration, SGN started using [Amazon AppStream 2.0](#), a secure, reliable, and scalable application streaming and low-cost virtual desktop service. Using a fully managed application streaming service helped the team work remotely and create a secure data store on the cloud instead of on-premises data centers. “It was critical for us to move off obsolete legacy infrastructure platforms and to a more evergreen world where we could benefit from cloud computing tools and services as standard,” says Smith.

Working alongside its AWS solutions architect and technical account manager, SGN deployed its Gas Control Governance Structure using [AWS Control Tower](#), which orchestrates multiple AWS services for an organization while maintaining security and compliance needs. SGN drafted a proof of concept and worked with its partners to spin up instances to test applications in a virtual environment. The proof of concept was completed in August 2019. After that, the company created logical and conceptual architecture designs to map out how to build the most secure environment. Those designs were finished by September 2020. Following that, the company ran a procurement exercise to engage with a delivery partner who could deliver against the approved designs. SGN could begin to build only after those were given the green light.

In November 2021, SGN started building the new solution. This process proved challenging because SGN discovered that many of the designs and environments from other vendors in the early phases did not match the standards set by the technical teams. As a result, the team needed to go through many iterations and continually refine and test.

In September 2022, the company reached out to [AWS Professional Services](#), a global team of experts that can help businesses realize desired business outcomes on AWS, to get feedback on building, testing, and remediation. SGN then chose to bring in AWS Professional Services as the primary infrastructure deployment partner. AWS Professional Services designed and provided automation and integration capabilities for the DevSecOps pipelines and tooling, as well as tooling and processes for certificate management, backup, and patching. The team also deployed and configured AWS native security and governance solutions. Finally, SGN went full-scale on the new solution in May 2023. “AWS Professional Services was integral in helping us implement the right solution to achieve our goals,” says Rajaram.

Outcome | Laying the Foundation for Future Energy Projects

Having successfully implemented the new infrastructure, SGN has now embraced a full-cloud approach using AWS. SGN plans to pursue future projects and opportunities, such as creating a gas network and green hydrogen digital twin. With its enhanced security stance, SGN is confident in the groundwork it has laid to continue evolving and innovating.

“The use of leading-edge AWS cloud computing technologies helps us stay one step ahead of the ever-changing cyber threat landscape,” says Smith.

About SGN

SGN Ltd. is a UK gas distribution company that manages natural and green gas networks, serving almost six million homes and businesses across Northern Ireland, Scotland, and the south of England.

AWS Services Used

Amazon AppStream 2.0

AppStream 2.0 is an AWS End User Computing (EUC) service that can be configured for SaaS application streaming or delivery of virtual desktops with selective persistence.

[Learn more »](#)

AWS Control Tower

AWS Control Tower orchestrates multiple AWS services on your behalf while maintaining the security and compliance needs of your organization.

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AWS Transit Gateway

AWS Transit Gateway connects your Amazon Virtual Private Clouds (VPCs) and on-premise networks through a central hub. This connection simplifies your network and puts an end to complex peering



relationships.

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AWS Professional Services

The AWS Professional Services organization is a global team of experts that can help you realize your desired business outcomes when using the AWS Cloud.

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